24. Monday, August 12th. Compact Metric Spaces

- If A (X, d) is bounded, we define $\operatorname{diam}(A) = \sup\{d(a,b) \mid a,b \in A\}$

The Lebesgue number Lemma: Let (X, d) be a compact metric space and let $\mathcal{U}$ be an open covering of X. Then there exists $\delta > 0$ such that for every subset $A \subseteq X$ with $\operatorname{diam}(A) < \delta$, there exists $U \in \mathcal{U}$ with $A \subseteq U$

Such δ is called a Lebesgue number for $\mathcal{U}$

Proof: Note first that if $X \in \mathcal{U}$, any δ will work, so assume $X \notin \mathcal{U}$.

Since $\mathcal{U}$ is an open cover and X is compact, there exist $U_1, \dots, U_n \in \mathcal{U}$ such that $X = U_1 \cup \dots \cup U_n$.

For each $1 \leq i \leq n$ consider the non-empty closed set $C_i = X \setminus U_i$.

Now define $f: X \to \mathbb{R}$ by

$$f(x) = \frac{1}{n} \sum_{i=1}^{n} d(x, C_i)$$

which is continuous because each map $X \to d(x, C_i)$ is continuous.

Claim 1: $f(x) > 0$ for all $x \in X$.

In fact, given $x \in X = U_1 \cup \dots \cup U_n$. Take j such that $x \in U_j$. Since U_j is open we have $B(x, \varepsilon) \subseteq U_j$ for some $\varepsilon > 0$. Hence $d(x, C_j) \geq \varepsilon$,

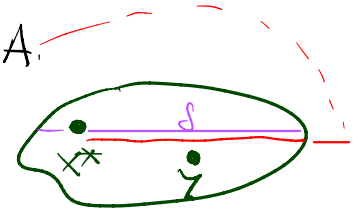
Therefore $n f(x) = \sum_{i=1}^{n} d(x, C_i) \geq d(x, C_j) \geq \varepsilon$

Thus $f(x) \geq \frac{\varepsilon}{n} > 0$ $\square_{\text{Claim 1}}$

Since $f: X \to \mathbb{R}_{>0}$ is continuous and X is compact, it has a minimum value $\delta = f(x_{min}) > 0$ such that $\delta \leq f(x) \quad \forall x \in X$

Claim 2: This δ serves as Lebesgue number for $\mathcal{U}$

Let $A \subseteq X$ bounded with $\operatorname{diam}(A) < \delta$, fix some $x^* \in A$.



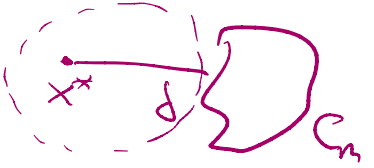
Step 1: $A \subseteq B(x^*, \delta)$

For every $y \in A$, we have $d(x^*, y) \leq \operatorname{diam}(A) < \delta$. Thus $A \subseteq B(x^*, \delta)$

Step 2: $B(x^*, \delta)$ is contained in some $U_1, \dots, U_n$.

Take m such that $d(x^*, C_m) = \max\{ d(x^*, C_i) \mid 1 \leq i \leq n \}$. For this m we have

$$\delta \leq f(x^*) = \frac{1}{n} \sum_{i=1}^{n} d(x^*, C_i) \leq \frac{1}{n} \sum_{i=1}^{n} d(x^*, C_m) = d(x^*, C_m)$$



Therefore $B(x^*, \delta) \subseteq U_m$ □ claim 2.

Hence $A \subseteq B(x^*, \delta) \subseteq U_m$, as desired □

Definition: Let (X, d_X) and (Y, d_Y) metric spaces. We say that a map $f: X \to Y$ is **uniformly continuous** if

Given $\varepsilon > 0$, $\exists \delta > 0$ such that, $\forall x \in X$, $f(B_X(x, \delta)) \subseteq B_Y(f(x), \varepsilon)$

Uniform Continuity Theorem: Let $f: (X, d_X) \to (Y, d_Y)$ be a continuous map. If X is compact, Then f is uniformly compact

Proof: Take $\varepsilon > 0$. Consider the open cover $\mathcal{U}' = \{ B_Y(y, \frac{\varepsilon}{2}) \mid y \in Y \}$ of Y.

Since f is continuous, $\mathcal{U} = \{ f^{-1}(B(y, \frac{\varepsilon}{2})) \mid y \in Y \}$ is an open cover of X.

Let δ be a Lebesgue number for $\mathcal{U}$, which exists because X is compact

We claim this $\delta > 0$ works. Take $x_1, x_2 \in X$ with $d_X(x_1, x_2) < \delta$. Then $\operatorname{diam}\{x_1, x_2\} < \delta$

Thus $\{x_1, x_2\} \subseteq f^{-1}(B(y, \frac{\varepsilon}{2}))$ for some $y \in Y$

Therefore $f(\{x_1, x_2\}) \subseteq B(y, \frac{\varepsilon}{2})$

Hence $d_Y(f(x_1), f(x_2)) \leq d_Y(f(x_1), y) + d_Y(y, f(x_2)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$

This means That f is uniformly continuous $\square$